

Mohammad Javad Najafirad

Work Authorization: Employment Authorization Document (EAD) Holder

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Summary

Final-year Ph.D. candidate in Electrical & Computer Engineering (expected May 2027) with nearly a decade of experience spanning hands-on electrical installation, power systems, power electronics, and advanced control. Specializes in dynamic modeling and distributed control of AC/DC microgrids and inverter-based resources, turning rigorous theory (LMI optimization, stability analysis) into practical solutions that improve grid reliability. Proficient in MATLAB/Simulink, Python, Typhoon HIL, and PSCAD.

Education

- Stevens Institute of Technology** New Jersey, USA
Ph.D. Electrical & Computer Engineering (GPA: 3.85/4.00) September 2023 – August 2027
Dissertation Title: "Dissipativity-Based Distributed Control Framework for AC/DC Microgrids"
- Shahid Rajaei University** Tehran, Iran
M.Sc. Electrical Control Engineering (GPA: 3.76/4.00) September 2019 – August 2022
Master's Thesis: "Distributed Event-Triggered Control with Input Saturation and Delay for DC Microgrids"
- Islamic Azad University** Tehran, Iran
B.Sc. Electrical Power Engineering September 2015 – September 2019
Bachelor's Project: "Power Factor Modification and Capacitor Banking Operation in Harmonic Networks"

Research Interests & Skills

Research Interests: Power Systems/Power Electronics, DC/AC Microgrids, Control System Design, Optimization

Relevant Coursework: Power Systems, Power Electronics, Modern Control, Optimal Control, Nonlinear Control, Multivariable Control, System Identification, Machine Learning

Modeling & Simulation: State-space modeling of power converters, microgrids, and BESS; stability analysis

Programming: MATLAB, Python, C, $\LaTeX$, Git, Arduino, Raspberry pi

Software: Simulink, Typhoon HIL, PSCAD, DIgSILENT, Altium Designer, Proteus, ETAP, Microsoft Office

Experience

Stevens Institute of Technology Hoboken, NJ
Research Assistant September 2023 – Present

- Designed droop-free distributed controllers for DC/AC microgrids with LMI-based stability guarantees, ensuring accurate voltage regulation and current sharing under ZIP loads, input saturation, and disturbances.
- Co-designed controller gains and communication topology via convex optimization, reducing communication links while preserving stability; outperformed droop and distributed secondary control.
- Developing a battery energy storage (BESS) framework for DC microgrids with distributed SoC balancing.
- Built MATLAB/Simulink and YALMIP/MOSEK tools for network modeling and LMI controller synthesis.

Electro Panelboard Company Tehran, Iran
Engineering Intern September 2018 – December 2018

- Assembly of high and low voltage power panels for residential and industrial applications.
- Wired, tested, and inspected switchgear and protection devices per single-line diagrams.

Self Employed Tehran, Iran
Electrician September 2016 – August 2018

- Implementation of electrical installations in residential buildings in Tehran province.
- Installed and troubleshot lighting, outlets, distribution boards, and grounding systems.

Top Publications (Google Scholar)

1. **M. J. Najafirad**, S. Welikala, "Dissipativity-Based Control and Communication Co-Design for DC Microgrids with Constant-Power Loads and Input Saturation," *2026 IEEE Conference on Decision and Control (CDC)*.
2. **M. J. Najafirad**, S. Welikala, L. Wu, P. J. Antsaklis, "Dissipativity-Based Distributed Control and Communication Topology Co-Design for Robust AC Microgrids," *2026 European Control Conference*, Reykjavik, Iceland.
3. **M. J. Najafirad** and S. Welikala, "Dissipativity-Based Distributed Control and Communication Topology Co-Design for DC Microgrids with ZIP Loads," *2026 IEEE American Control Conference*, New Orleans, LA, USA.
4. **M. J. Najafirad** and S. Welikala, "Dissipativity-Based Distributed Droop-Free Controller and Communication Topology Co-Design for DC Microgrids," *2025 IEEE American Control Conference*, Denver, CO, USA.
5. **M. J. Najafirad**, N. M. Dehkordi, M. Hamzeh, and H. Nazaripouya, "Distributed Event-Triggered Control of DC Microgrids With Input Saturation and Time Delay Constraints," *IEEE Systems Journal*, 2023.

Academic Projects

Distributed Control of DC Microgrids

January 2021 – September 2022

- Developed voltage restoration algorithms with precise power sharing through pinning gain control.
- Developed event-triggered algorithms to reduce communication among renewable energy resources.
- Implemented a hybrid control architecture: continuous-time control and discrete-time control.
- Designed a supervisor mechanism for dynamic controller switching, optimizing performance and stability.

VSG Control of VSC-MTDC Systems

- Designed a parameter-alternating virtual synchronous generator (VSG) controller for multi-terminal HVDC.
- Damped low-frequency oscillations by adapting the virtual inertia and damping parameters in real time.

Machine Learning Facial Recognition System

December 2024

- Developed ML models to classify facial expressions and attributes across demographics.
- Implemented a custom three-layer CNN (ReLU, Adam optimizer) for robust feature extraction.

Honors and Awards

- Best Paper Award Nominee, IEEE American Control Conference (ACC) July 2025
- Recipient of Full Academic Fellowship, Stevens Institute of Technology September 2023
- Ranked 2nd GPA among M.Sc. students, Shahid Rajaee University August 2022
- Top-ranked in Persian National University Entrance Exams August 2015 & August 2019

Volunteer Activities

- President, Iranian Association at Stevens Institute of Technology May 2026-Present
- Reviewer, IEEE Power & Energy Society March 2025-Present
- PhD Liaison, Graduate Student Council at Stevens Institute of Technology 2025-2027
- Senior Executive in IEEE PEDST Conference February 4-6, 2020
- Executive Staff of IEEE Student Association at IAU South Tehran Branch October 2017 - May 2019
- Translator of TED Talks into Persian September 2017 - Present

Languages

English: Fluent – TOEFL iBT 100/120

Persian: Native

Spanish: Elementary

References

Dr. Shirantha Welikala: Assistant Professor, Stevens Institute of Technology
Dr. Hamidreza Nazaripouya: Assistant Professor, Oklahoma State University
Dr. Nima Mahdian Dehkordi: Associate Professor, Shahid Rajaee University
Dr. Mohsen Hamzeh: Associate Professor, University of Tehran

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